

BLUESTOCK MUTUAL FUND ANALYTICS

Capstone Project

TOOLS USED:
PYTHON | PANDAS | JUPYTER | TABLEAU |
GITHUB

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Problem Statement

Mutual fund datasets contain large volumes of NAV, transaction, and performance data. Investors and analysts require an integrated platform to evaluate fund performance, investor behavior, and market trends using data-driven insights.

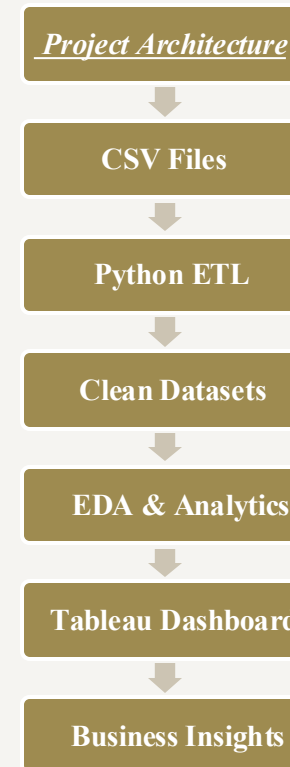
Objective:

Build an end-to-end analytics solution for mutual fund analysis, risk evaluation, investor analytics, and dashboard reporting.

Data Sources and Project Architecture



- **clean_nav.csv**
- **clean_transactions.csv**
- **clean_performance.csv**



EDA Highlights

- Small Cap funds showed higher returns.
- SIP transactions dominated investor activity.
- Direct plans had lower expense ratios.

Key Findings

- Small Cap funds delivered the highest returns but with greater volatility.
- SIP transactions accounted for the majority of investor activity.
- Direct plans generally had lower expense ratios than regular plans.
- Mutual fund participation increased steadily across the analysis period.
- AUM was concentrated among a few leading fund houses.

Observations

- *Investment activity showed a positive long-term trend.*
- *Fund performance varied significantly across categories.*
- *Retail investor participation remained strong through SIP investments.*
- *Market growth was reflected in increasing assets under management.*

Performance Analytics

<u>Metric</u>	<u>Purpose</u>
CAGR	Annualized growth measurement
Sharpe Ratio	Risk-adjusted return
Sortino Ratio	Downside risk-adjusted return
Alpha	Excess return over benchmark
Beta	Market sensitivity
Max Drawdown	Largest historical decline

Advanced Analytics

Risk & Investor Analytics

Risk Analytics

- Historical VaR (95%)
- Conditional VaR (CVaR)
- Rolling 90-Day Sharpe Ratio

Investor Analytics

- Cohort Analysis
- SIP Continuity Analysis
- At-Risk Investor Identification

Fund Recommendation Engine

- Built using Sharpe Ratio and Risk Grade
- Recommends top funds based on investor risk appetite

Business Value

- Improves investment decision-making
- Helps identify investor retention risks
- Supports personalized fund recommendations

Dashboard Overview

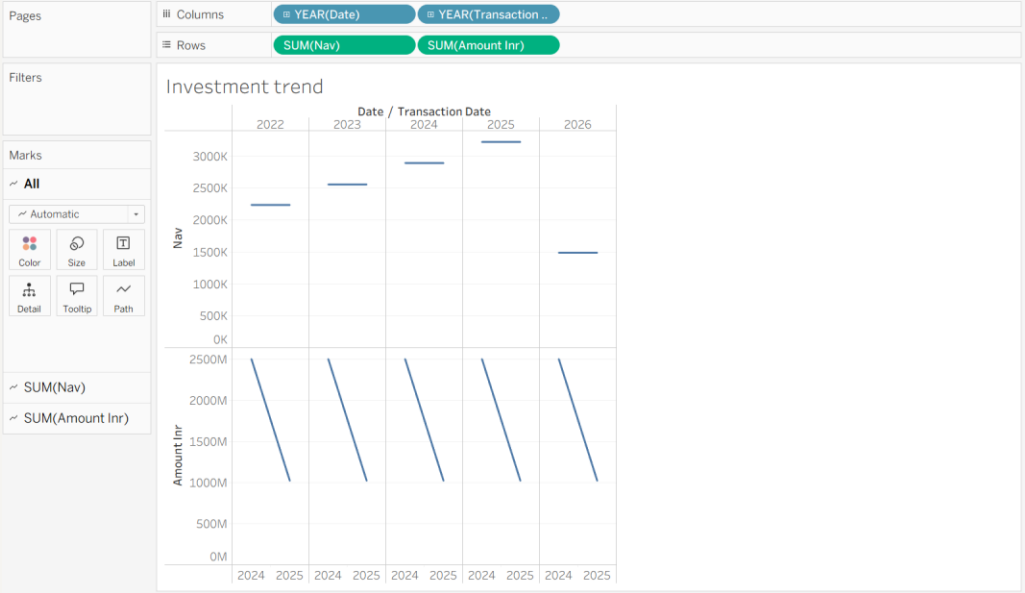
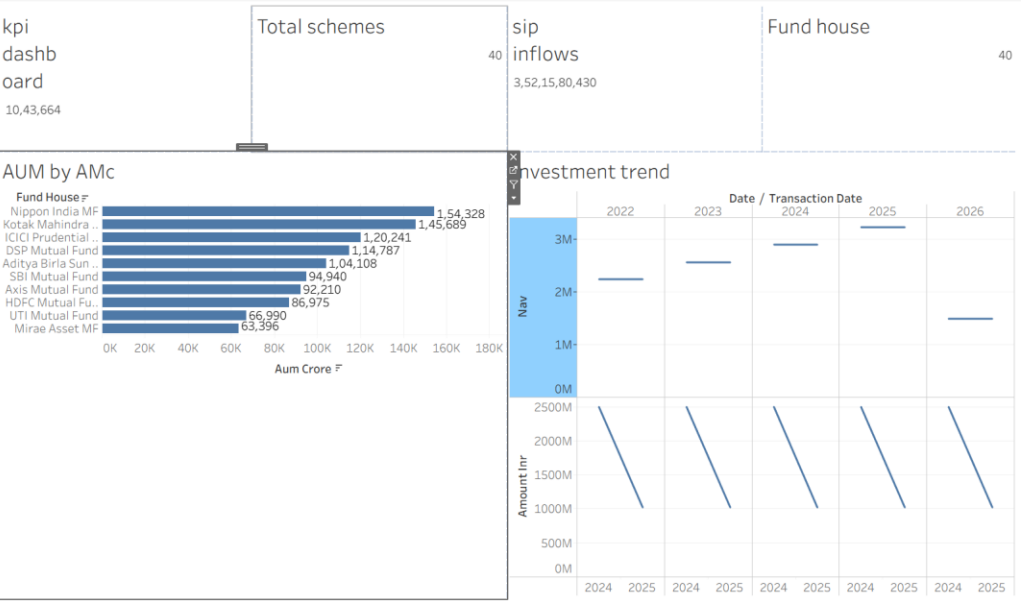
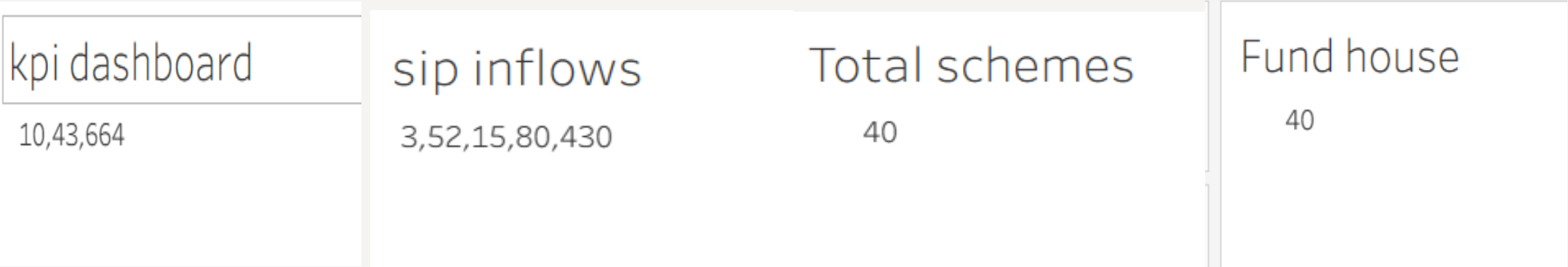
Tableau Dashboard Features

- KPI Monitoring
- Fund House Comparison
- SIP Inflow Analysis
- Investment Trend Tracking
- Scheme Distribution Analysis

Benefits

- Interactive filtering
- Easy comparison of fund performance
- Quick identification of market trends

Dashboard Visualizations



Dashboard Insights

Major Insights Generated

- SIP participation remained the dominant investment mode.
- Leading AMCs controlled a significant portion of total AUM.
- Investment trends showed sustained market growth.
- Risk-adjusted metrics highlighted high-performing schemes.
- Investor participation continued to expand across categories

Recommendations

For Investors

- Focus on risk-adjusted returns rather than returns alone.
- Maintain SIP continuity for long-term wealth creation.
- Diversify investments across fund categories.

For Asset Management Companies

- Improve investor engagement to reduce SIP discontinuation.
- Promote direct plans for cost-conscious investors.
- Use analytics-driven insights for product positioning.

Future Enhancements

- Real-time market data integration
- Machine Learning-based fund recommendations
- Predictive investor churn analysis

Conclusion & Thank You

Project Conclusion

- Successfully built an end-to-end Mutual Fund Analytics solution.
- Performed ETL, EDA, performance analysis, and advanced risk analytics.
- Developed interactive Tableau dashboards for business reporting.
- Generated actionable insights into fund performance and investor behavior.

Skills Applied

Skills Applied

- Python
- Pandas
- Data Analytics
- Tableau
- Business Intelligence
- GitHub Version Control

Thank You